

NLD Module - Stage 3 Functional Process Document

Objective

Stage 3 is the **Network Design (NLD) Stage** where the engineer finalizes the end-to-end topology based on the feasibility received from Stage 2.

The system allows the engineer to:

- Review Stage 2 feasibility results.
 - Select the final A End and B End.
 - Decide whether each end is Existing or New.
 - Configure the required handoff.
 - Automatically generate dependent POP Feasibility and On-Net/Off-Net requests whenever a new POP is required.
 - Maintain complete parent-child relationships between all generated requests.
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Step 1 - Display Stage 2 Results

When Stage 3 is opened, display all Stage 2 records.

Each record must display its current status:

- Feasible
- Not Feasible
- Information Awaited

Only **Feasible** records can proceed for Network Design.

Step 2 - Display End Details

A End

Display the A End details received from Stage 2.

B End

Display all feasible B End options received from Stage 2.

Multiple B Ends may exist.

The engineer can select **only one** B End for further processing.

Step 3 - Validate End Type

After selecting A End and B End, determine whether each end is:

- Existing POP
- New L1/L2 POP

Possible combinations:

A End	B End	Action
Existing	Existing	Continue normal NLD flow
New	Existing	Configure A End
Existing	New	Configure B End
New	New	Configure both independently

Both A End and B End may be New simultaneously.

Each New End is processed independently.

Step 4 - Existing POP Scenario

If both A End and B End are Existing POPs:

- No additional POP creation is required.
 - No nested requests are generated.
 - Continue to the remaining NLD workflow.
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Step 5 - New POP Configuration

Whenever an End is selected as **New L1/L2 POP**, additional configuration is required.

For each New End, display:

Handoff Type

- Direct
 - On Net
 - Off Net
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Step 6 - Direct Handoff

If Direct is selected:

Display:

- Parent POP

The engineer selects one existing Parent POP.

Purpose:

Although this is a Direct NLD Handoff, the newly created POP must still be logically connected to an existing Parent POP within the network topology.

Example:

Parent POP

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New NLD POP

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Job POP (Final Endpoint)

No On-Net or Off-Net request is generated.

No additional transport selection is required.

Step 7 - On Net Handoff

If On Net is selected:

Display:

- Parent POP

Then display Transport Type:

- Wireless
- Wireline
- Ring

After submission, the system automatically creates:

- New POP Feasibility Request
- New On-Net Request

These requests are linked with:

- Original NLD Job
- Original POP Request
- Current NLD Request

The generated requests follow the predefined On-Net workflow.

Step 8 - Off Net Handoff

If Off Net is selected:

Display:

- Parent POP

Display Transport Type:

- Wireless
- Wireline
- Ring

After submission, automatically create:

- New POP Feasibility Request
- New Off-Net Request

These requests follow the existing Off-Net workflow.

Step 9 - Independent Processing of A End and B End

A End and B End are processed independently.

Examples:

Scenario 1

A End = Existing

B End = Existing

Result:

Normal NLD processing.

Scenario 2

A End = New (Direct)

B End = Existing

Result:

Generate only A End topology.

Scenario 3

A End = Existing

B End = On Net

Result:

Generate only B End nested requests.

Scenario 4

A End = On Net

B End = Off Net

Result:

Generate nested requests for both ends independently.

Step 10 - Network Topology

The Job POP always remains the **final endpoint**.

It never becomes an intermediate routing node.

Examples:

Existing to Existing

Existing POP



Job POP

New A End (Direct)

Parent POP



New A POP



Job POP

New B End (Direct)

Job POP



New B POP



Parent POP

New A End + New B End

Parent POP (A)



New A POP



Job POP

↓

New B POP

↓

Parent POP (B)

The Job POP remains the central endpoint of the original Job Request.

It is never replaced by generated POPs.

Step 11 - Automatic Request Generation

Whenever On Net or Off Net is selected, the system automatically creates:

POP Feasibility Request

Contains:

- Original Job Reference
 - Original POP Request Reference
 - Parent NLD Request Reference
 - Generated POP Reference
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On Net / Off Net Request

Contains:

- Generated POP Reference
- Parent POP
- Selected Transport
- Original Job Reference
- Original NLD Reference

These requests are treated as dependent child requests and do not follow the complete original NLD lifecycle.

Step 12 - Reference Mapping

Every generated request must maintain complete traceability.

Store references for:

- Original Job ID
- Original POP Request ID
- Parent NLD Request ID
- Parent POP
- Child POP
- Generated POP Request
- Generated On-Net Request
- Generated Off-Net Request
- Transport Type
- Handoff Type

This ensures complete hierarchy tracking.

Step 13 - Business Rules

Rule 1

Only one B End can be selected.

Rule 2

Only Feasible Stage 2 records can proceed.

Rule 3

Both A End and B End may independently be Existing or New.

Rule 4

Each New End is processed separately.

Rule 5

Direct Handoff requires Parent POP selection.

Rule 6

On Net and Off Net always require:

- Parent POP
 - Transport Type
-

Rule 7

On Net automatically generates:

- POP Feasibility Request
 - On Net Request
-

Rule 8

Off Net automatically generates:

- POP Feasibility Request
 - Off Net Request
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Rule 9

Generated requests are linked to the original Job and original POP Request.

Rule 10

The Job POP always remains the final endpoint.

It never changes into an intermediate routing node.

Rule 11

There is no recursive nesting.

Generated POP requests cannot continue generating additional levels indefinitely.

Stage 3 Workflow Summary

Stage 2 Feasible Records

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Select Feasible Record

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Select One B End

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Identify Existing/New for A End

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Identify Existing/New for B End

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If Existing

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Continue NLD

OR

If New

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Select Handoff Type

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Direct

OR

On Net

OR

Off Net

↓

Select Parent POP

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(If On Net/Off Net)

Select Transport

↓

Automatically Generate Child Requests

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Maintain Parent-Child References

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Continue Remaining NLD Workflow